

Penetration Testing Report

Vulnhub – DC1

Executive Report

Target

Linux - Drupal

Assesment Type

Penetration Testing Laboratory

Prepared by

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Assessment Period

June 2026

CONFIDENTIALITY

Educational & Professional Portfolio

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1. Statement of confidentiality

This report was produced as a part of a personal penetration testing laboratory exercise.

The information contained in this document is intended solely for a educational and professional portfolio purposes. Any IP addresses , credential, flag and system information referenced herein belong to the target laboratory environment and must not be used against unauthorized systems.

Distribution of this report should be limited to educational, recruitment and professional development purposes.

2. Engagement contacts

Assessor Contact			
Assessor name	Title	Project	Assessor email
X	Project Director	DC-1	X@vlncompany.vln
José Martín	Penetration Tester	DC-1	Jose@vlncompany.vln

3. Executive Summary

A penetration test was conducted againsts the DC-1 virtual machine from Vulnhub.

The objective of the assessment was to identify vulnerabilities that could allow unathorized access to the target system and evaluate the impact of a successful compromise.

The test demonstrated weaknesses in web application security, credential managment and operating system hardening.

4. Approach

The assessment followed a standard penetration testing methodology:

1. Reconnaissance
2. Service enumeration
3. Web application enumeration
4. Vulnerability research
5. Initial - Exploitation
6. Post - Exploitation
7. Privilege escalation

8. Evidencie coleccion

Each phase was documented and validated through technical evidence.

5. Assignment

Assignment			
From	To	Start	End
Project Director	José Martín	17/06/2026	22/06/2026

Target		
Name	Platform	Operating System
DC - 1	Vulnhub	Linux

The assignment`s objective are

- Obtain initial access to the target system
- Escalate privileges to root
- Recover all available flags
- Document the attack path and identified vulnerabilities

6. Scope

In scope

- Linux operating system
- Web application
- MySQL Database
- Local privilege escalation vectors

7. Assessment summary

Assesment summary	
Findings	Severity
Drupalgeddon RCE	Critical
Credentials in Settings.php	High
Sensitive database information	Medium
SUID Misconfiguration	Critical

8. Detailed walkthrough

The process had the following phases

8.1 Reconnaissance

Network reconnaissance identified three exposed services:

- SSH
- HTTP
- RPCBind

Further HTTP reconnaissance revealed that the target was running Drupal 7.

8.2 Initial - Exploitation

Research into known Drupal vulnerabilities identified Drupalgeddon as a viable attack vector. A Metasploit implementation of the exploit was configured and executed successfully, resulting in a Meterpreter session on the target system.

8.3 Post - Exploitation

Enumeration of the web server directories revealed the presence of the Flag1.txt file. The content of the file led to discover the Drupal configuration file settings.php which not only contains the Flag2, also contained valid database credentials.

Using these credentials, access to the MySQL database was obtained and sensitive information, including Flag3, was recovered.

8.4 Privilege Escalation

Privilege escalation opportunities were investigated through the SUID enumeration.

A misconfigured SUID-enabled instance of /usr/bin/find was identified and abused to execute a privilege shell. This resulted in full root compromise of the target system.

9. Collected evidences

Evidence files collected during the assessment:

- whoami.txt
- id.txt
- Flag1.txt
- Settings.php
- Flag3.txt
- SUID_Files.txt
- evidencia_whoami.txt
- evidencia_id.txt
- evidencia_shadow.txt
- Flag4.txt
- finalflag.txt

10. Remediation summary

The vulnerabilities found that are recommended to be fixed are:

1° Drupalgeddon

Upgrade Drupal to a supported version.

2° Credentials

Remove plaintext credentials from the configuration files

3° SUID

Review and restrict SUID permissions on system binaries.

11. Appendix A: Finding severities

Each finding has been assigned a severity rating of critical, high, medium, low. The rating is based off of an assessment of the priority which each finding should be viewed and the potential each one has on the confidentiality, integrity and availability of DC-1's data.

Severity Definitions	
Rating	Severity rating definition
Critical	It allows a full compromise of the system or unauthorized access to highly sensitive information with little or no prior preparation. It can lead to a remote code execution, administrative access or complete control of the target system.
High	Allows significant access to sensitive data or functionality, privilege escalation or lateral movement. Requires an additional condition or has lesser impact than a critical vulnerability.
Medium	It may facilitate subsequent attacks or expose information that is useful to an attacker. The impact on integrity, confidentiality and availability of the data is moderate and typically requires chaining with other vulnerabilities to achieve a full exploitation.
Low	It has a limited impact or is primarily informational. It does not typically grant direct access to the system, but it can assist an attacker during the reconnaissance or enumeration phases.

12. Appendix B : Tools

The tools used for the assessment were

- Nmap
- Curl
- Whatweb
- Gobuster
- Searchsploit
- Metasploit
- MySQL client

13. Appendix C : Technical Report

See the attached file:

- DC-1_Technical_Report.odt